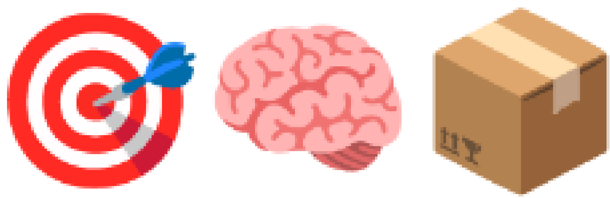
Defining the Problem

Engineering — Design Process | Unit 1, Lesson 1 of 6

📚 **Grade:** 3rd Grade ⏱️ **Duration:** 45–60 minutes 🛠️ **Subject:** Engineering — Design Process



Engineers begin by naming the need, the user, and what success should look like.

📋 **NGSS Standard:**
3-5-ETS1-1 — Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.

🎯 Learning Objective

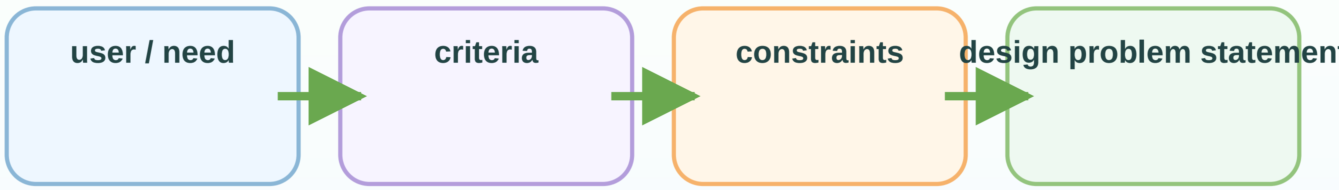
Students will define a grade-appropriate design problem by naming the user, the need, and clear success criteria for a fragile-object protection challenge.

💡 The Big Idea

A strong design starts before building. Engineers first decide who needs help, what problem needs to be solved, and what a successful solution must do.

🔬 The Science

SCIENCE SNAPSHOT



Specific design problems lead to better plans, better tests, and clearer evidence.

What is happening?

Engineers begin by naming the need, the user, and what success must look like. In this unit, a fragile-object protection challenge becomes stronger when students clearly say what must be protected and under what test conditions.

Why it works that way

A design problem needs criteria for success and constraints such as materials, time, or drop height. If those pieces are vague, later designs cannot be compared fairly because nobody agrees on what counts as success.

Look for this in the lesson

Students should focus on writing a precise problem statement before thinking about fancy builds. The clarity of the problem is what gives the whole unit structure.

Ask while you read: What exactly must the design do, and what limits must students respect?

Researching Constraints and Materials

Engineering — Design Process | Unit 1, Lesson 2 of 6

📚 **Grade:** 3rd Grade ⏱️ **Duration:** 45–60 minutes 🛠️ **Subject:** Engineering — Design Process

Engineers study their limits and materials before they choose a design.

📋 **NGSS Standard:**
3-5-ETS1-1 — Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.

🎯 Learning Objective

Students will identify important constraints and compare material properties for the fragile-object protection challenge.

💡 The Big Idea

Good engineering choices depend on limits. Engineers must understand the allowed materials, time, size, and cost before they decide what to build.

🔬 The Science

SCIENCE SNAPSHOT

Material properties

- soft / cushioning
- stiff / supports shape
- light or heavy

compare

Design needs

- absorb impact
- hold object
- survive drop test

Material research helps turn guessing into informed design choices.

What is happening?

Before building, engineers compare materials and learn the limits of the challenge. A material is useful only when its properties match the job the design needs it to do.

Why it works that way

For a fragile-object challenge, cushioning, stiffness, lightness, and shape all matter. Researching materials helps students predict which ones may absorb impact, spread force, or hold the object in place before they start building.

Look for this in the lesson

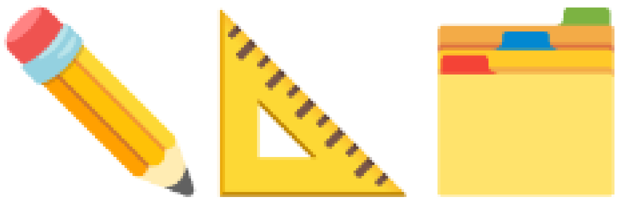
Students should talk about why a material might help. The lesson gets stronger when they compare properties instead of just naming favorite craft supplies.

Ask while you read: Which material property seems most important for protecting a fragile object?

Planning a Solution

Engineering — Design Process | Unit 1, Lesson 3 of 6

📚 **Grade:** 3rd Grade ⏱️ **Duration:** 45–60 minutes 🛠️ **Subject:** Engineering — Design Process



Engineers compare more than one plan before choosing what to build.

📋 **NGSS Standard:**
3-5-ETS1-2 — Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem.

🎯 Learning Objective

Students will sketch and compare at least two possible protective designs before choosing one to build.

💡 The Big Idea

Planning helps engineers test ideas on paper first. Comparing more than one possible solution leads to smarter building choices.

🔬 The Science

SCIENCE SNAPSHOT

sketch plan → label feature jobs → predict how it works → prepare to test

Plans are stronger when every feature is tied to a reason.

What is happening?

A plan is more than a picture. It should show how the parts of the design fit together and how the design is expected to solve the problem.

Why it works that way

Planning matters because it lets students think through force, support, and protection before they use materials. A clear plan also makes it easier to compare the original idea with what actually happened during testing.

Look for this in the lesson

Students should label each feature and explain its job. That makes the plan a scientific prediction rather than a decorative sketch.

Ask while you read: What feature in your plan is supposed to protect the fragile object most directly?

Building and Testing a Prototype

Engineering — Design Process | Unit 1, Lesson 4 of 6

📚 **Grade:** 3rd Grade ⏱️ **Duration:** 45–60 minutes 🛠️ **Subject:** Engineering — Design Process

A prototype lets engineers test ideas, collect evidence, and notice failure points.

📋 **NGSS Standard:**
3-5-ETS1-3 — Plan and carry out fair tests in which variables are controlled and failure points are considered to identify aspects of a model or prototype that can be improved.

🎯 Learning Objective

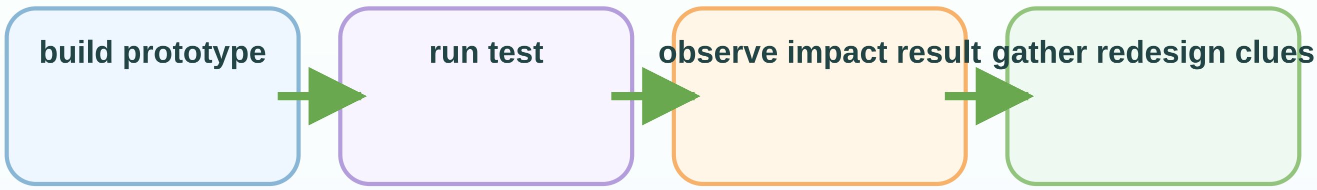
Students will build a prototype, test it fairly, and record evidence about strengths and failure points.

💡 The Big Idea

Testing turns a plan into evidence. A prototype helps engineers see what really happens instead of guessing.

🔬 The Science

SCIENCE SNAPSHOT



Prototype tests reveal how the design behaves under real challenge conditions.

What is happening?

A prototype is a first working version built so it can be tested. Engineers use prototypes to gather evidence about what works and what still needs improvement.

Why it works that way

During a drop test, the design may cushion, bounce, tip, crack, or fail in surprising ways. Those results are useful because they reveal how force moved through the system during impact.

Look for this in the lesson

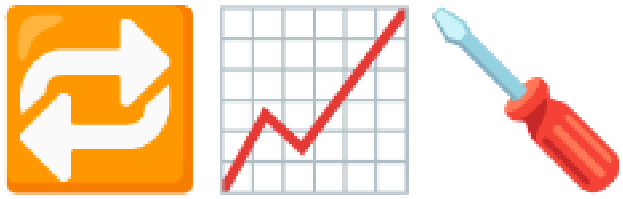
Students should record what happened to the design and to the fragile object. The goal is to identify which part of the system protected well and which part needs redesign.

Ask while you read: What happened during the test that tells you where the design succeeded or failed?

Improving the Design

Engineering — Design Process | Unit 1, Lesson 5 of 6

📚 **Grade:** 3rd Grade ⏱️ **Duration:** 45–60 minutes 🛠️ **Subject:** Engineering — Design Process



Engineers redesign by changing the part that evidence shows needs improvement.

📋 **NGSS Standard:**
3-5-ETS1-3 — Plan and carry out fair tests in which variables are controlled and failure points are considered to identify aspects of a model or prototype that can be improved.

🎯 Learning Objective

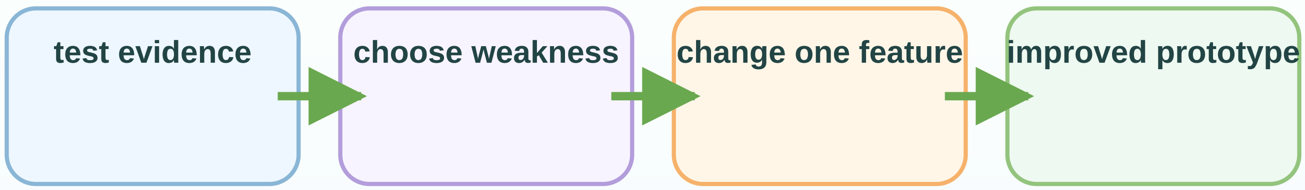
Students will revise version 1 into version 2 by making an evidence-based improvement and predicting how it should work better.

💡 The Big Idea

Redesign is not starting over randomly. Engineers use evidence from testing to decide what should change in the next version.

🔬 The Science

SCIENCE SNAPSHOT



The best redesigns are targeted responses to evidence, not random changes.

What is happening?

Improvement means using evidence from the first test to make the design more effective. Engineers rarely expect the first version to be the final version.

Why it works that way

A smart improvement targets a specific weakness. If the object moved around too much, the design may need a better holder. If the structure collapsed, it may need more support or a different shape.

Look for this in the lesson

Students should connect each revision to one observed problem. That makes redesign logical instead of random.

Ask while you read: Which observed weakness should you fix first, and why?

Share and Defend Your Solution

Engineering — Design Process | Unit 1, Lesson 6 of 6

📚 **Grade:** 3rd Grade ⏱️ **Duration:** 45–60 minutes 🛠️ **Subject:** Engineering — Design Process

Engineers defend a solution by connecting the problem, the design, and the evidence.

📋 **NGSS Standard:**
3-5-ETS1-2 — Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem.

🎯 Learning Objective

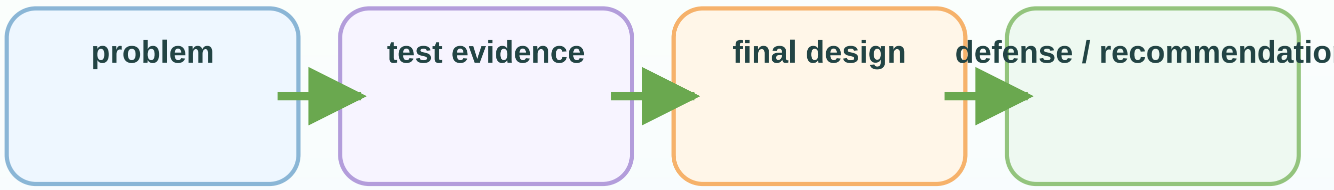
Students will present a final engineering solution and defend it with evidence from planning, testing, and redesign.

💡 The Big Idea

A strong final defense explains how the solution met the challenge criteria, followed the constraints, and improved because of evidence.

🔬 The Science

SCIENCE SNAPSHOT



A final defense turns building results into a reasoned engineering recommendation.

What is happening?

Engineers explain why their design is a good solution by connecting the challenge, the evidence from testing, and the final design choices. Communication is part of the engineering process, not an extra step.

Why it works that way

A strong defense shows how the design met criteria, handled constraints, and improved after testing. Evidence matters more than confidence; the best explanation points to what actually happened in the tests.

Look for this in the lesson

Students should compare their final design to an earlier version or an alternative. That comparison helps them explain why the final version is the best fit.

Ask while you read: What evidence best proves your final design is stronger than your earlier idea?